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Noh et al.

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(54) **SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE SAME**

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H01L 23/522 (2006.01)
H01L 23/532 (2006.01)

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(58) **Field of Classification Search**
CPC H01L 21/76877; H01L 21/76844; H01L 21/76843; H01L 21/76829
See application file for complete search history.

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Primary Examiner — Wael M Fahmy

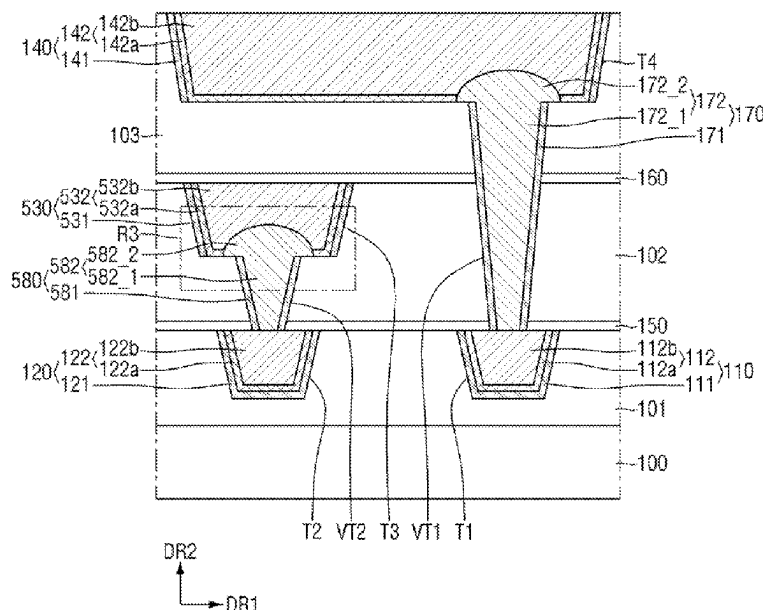
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(57) **ABSTRACT**

A semiconductor device includes: a substrate; a first interlayer insulating layer on the substrate; a first wiring pattern in a first trench of the first interlayer insulating layer; a second interlayer insulating layer on the first interlayer insulating layer; a second wiring pattern in a second trench of the second interlayer insulating layer; a third interlayer insulating layer on the second interlayer insulating layer; a third wiring pattern in a third trench of the third interlayer insulating layer, and including a wiring barrier layer and a wiring filling layer, wherein the wiring filling layer contacts the third interlayer insulating layer; a via trench extending from the first wiring pattern to the third trench; and a via including a via barrier layer and a via filling layer. The via barrier layer is in the via trench. The via filling layer contacts the first wiring pattern and the wiring filling layer.

16 Claims, 17 Drawing Sheets



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FIG. 2

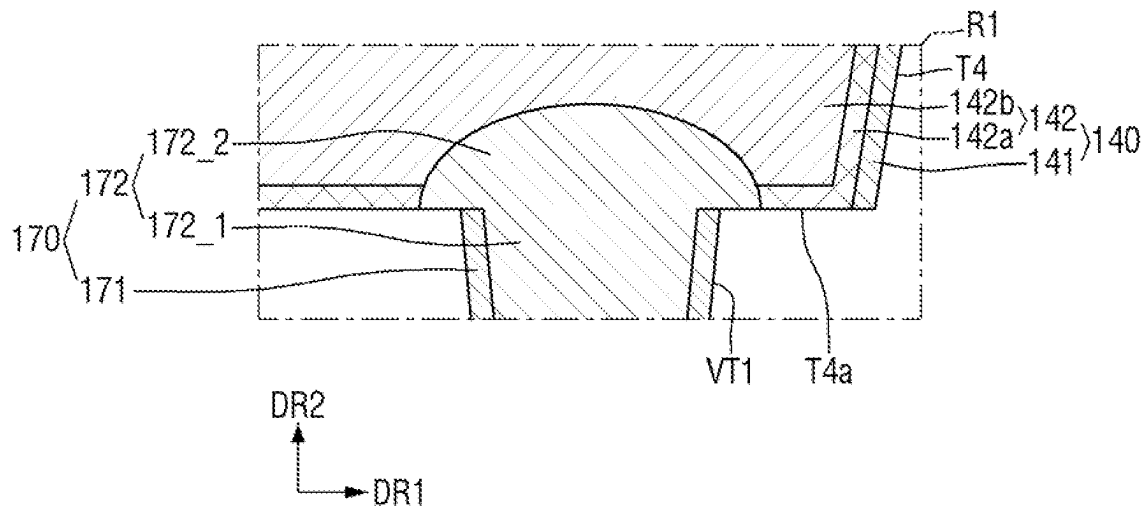


FIG. 3

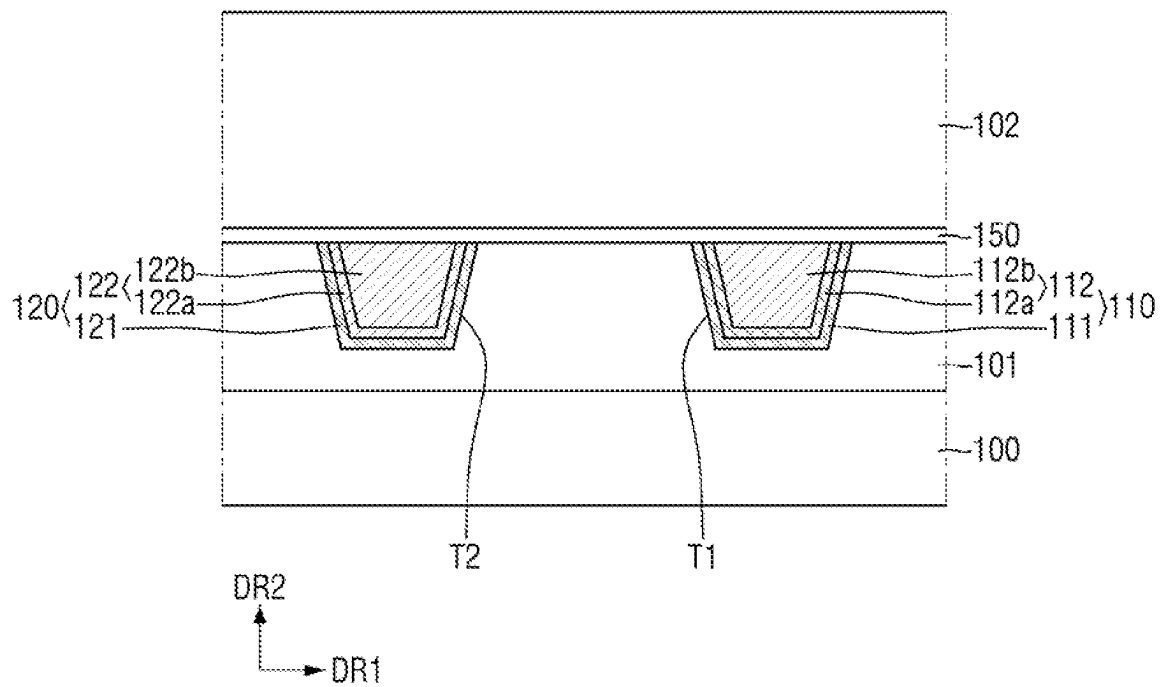


FIG. 4

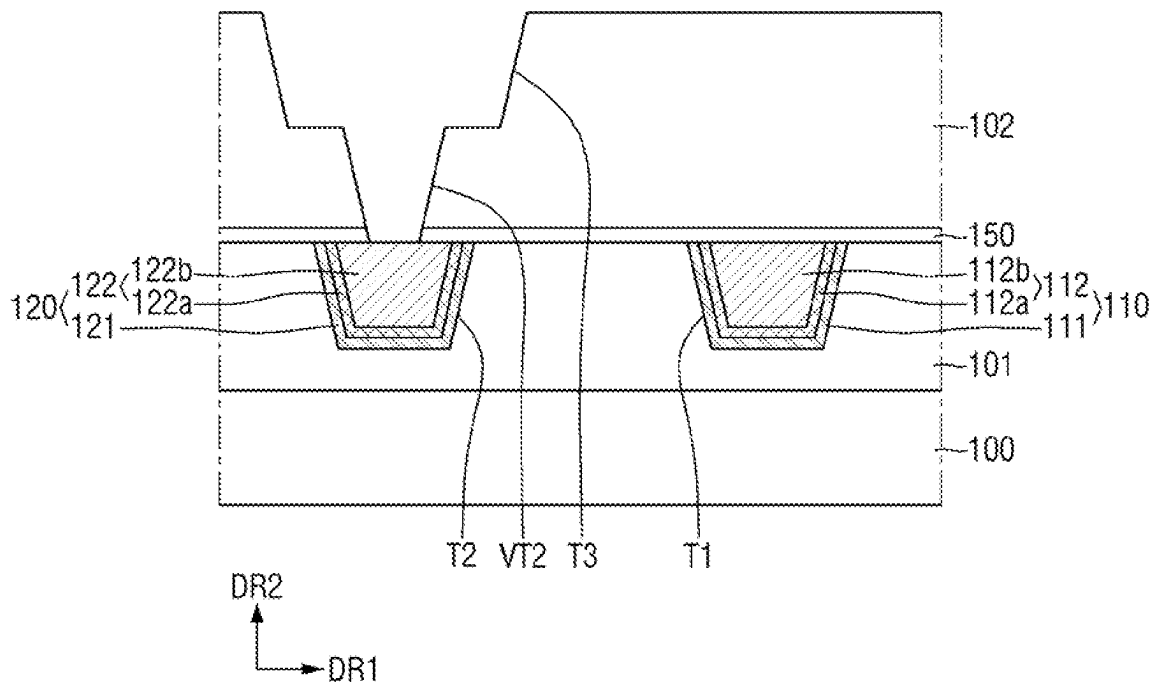


FIG. 5

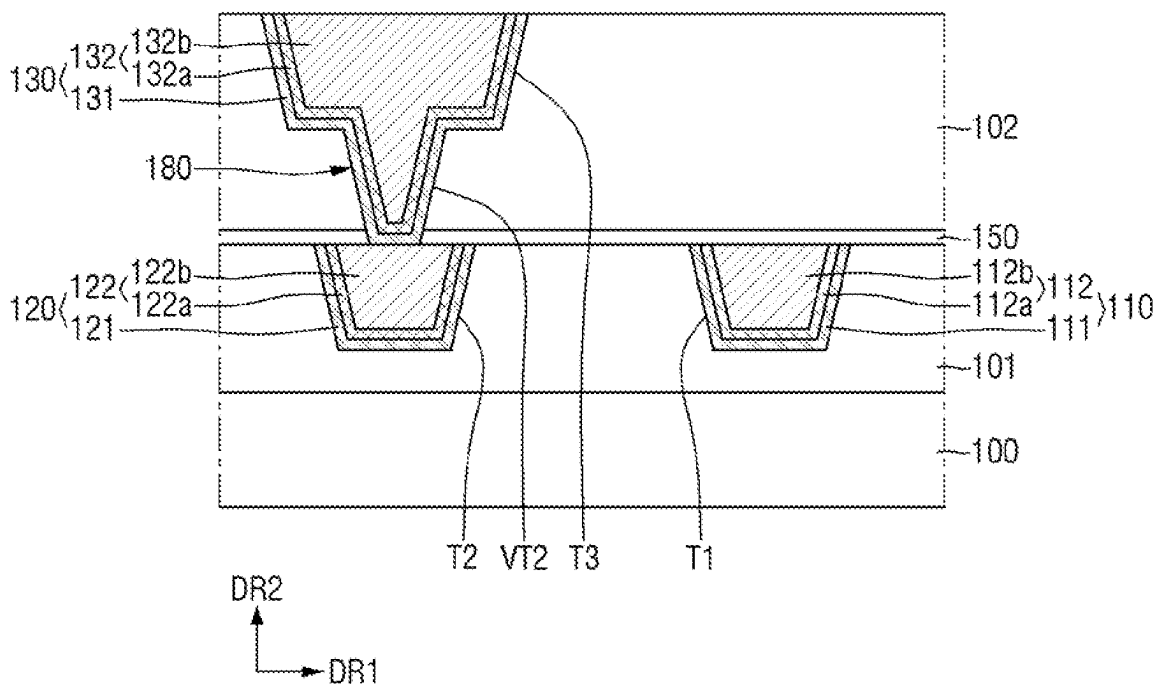


FIG. 6

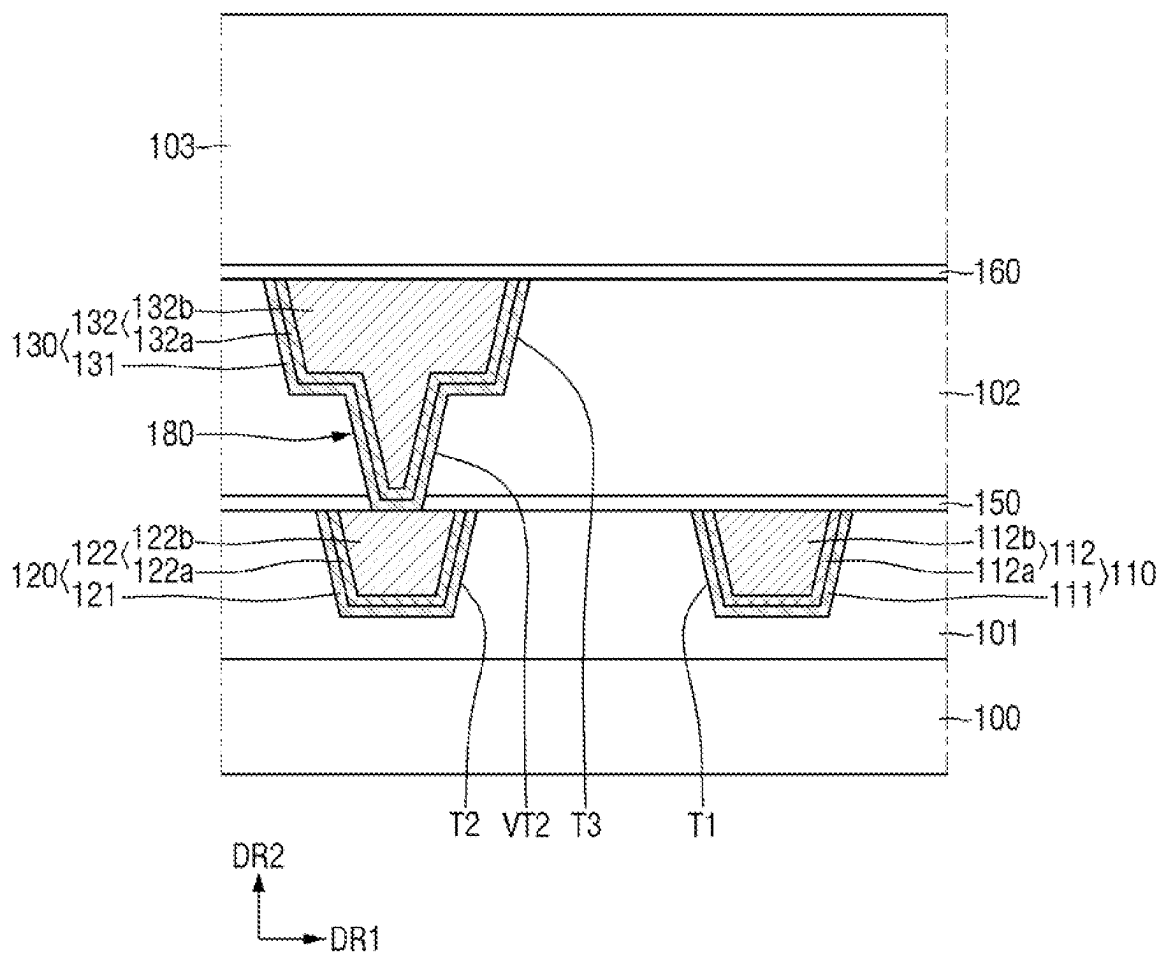


FIG. 7

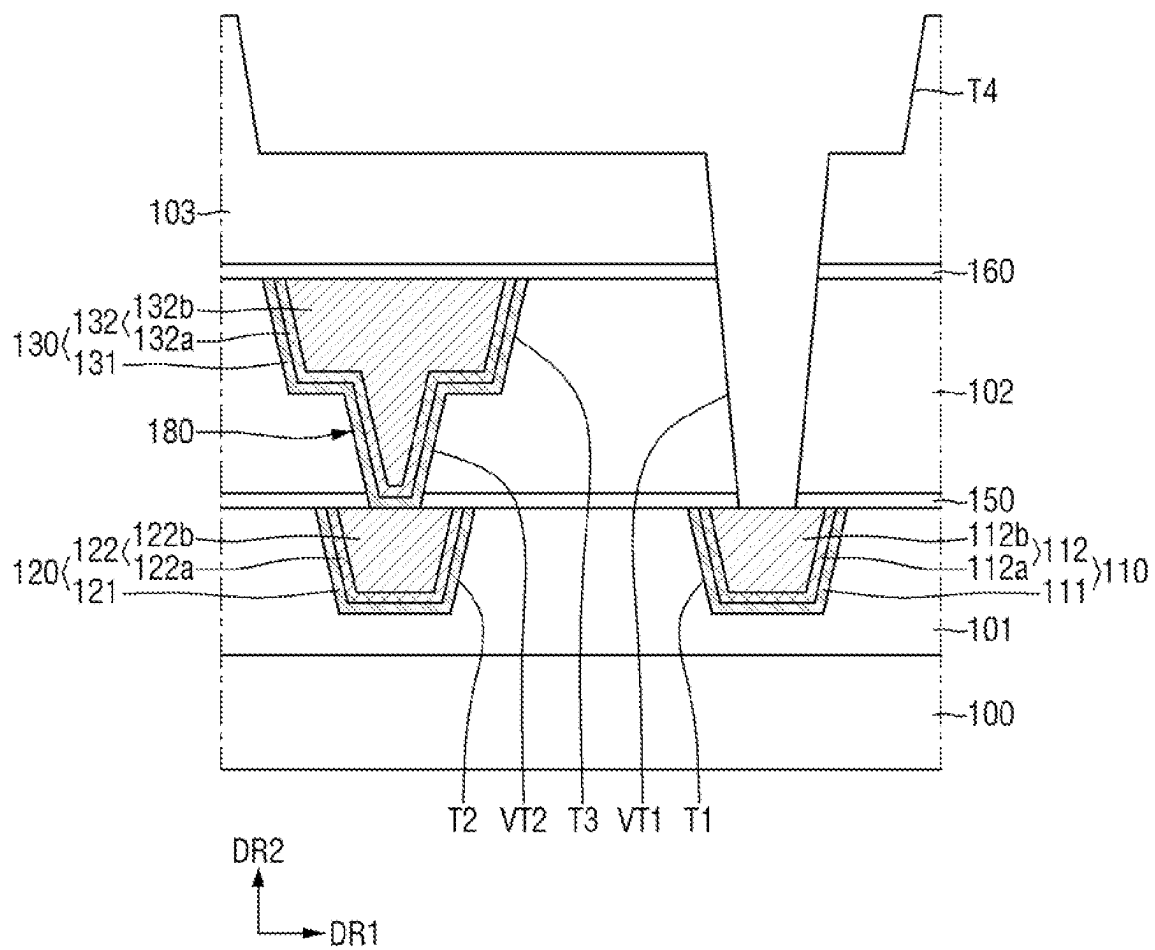


FIG. 8

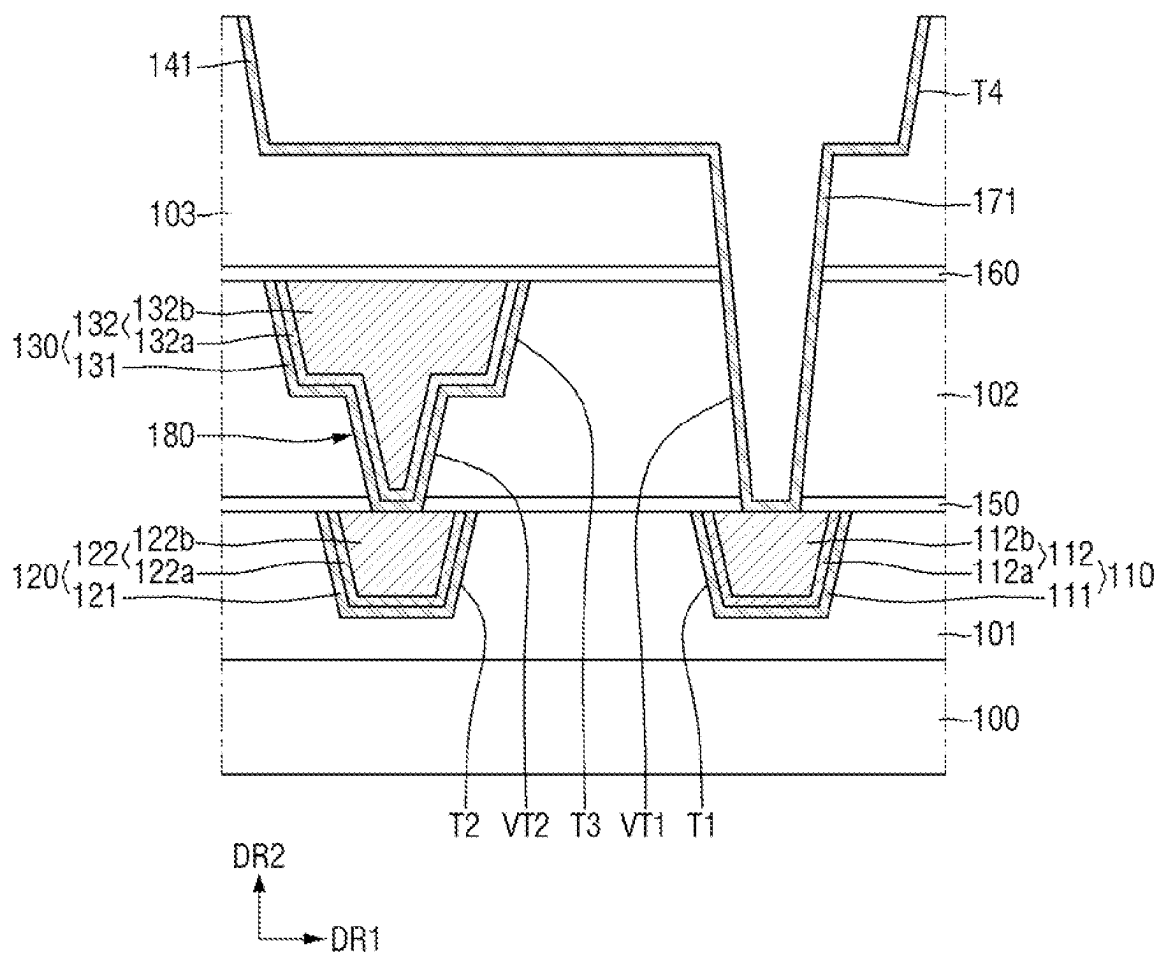


FIG. 9

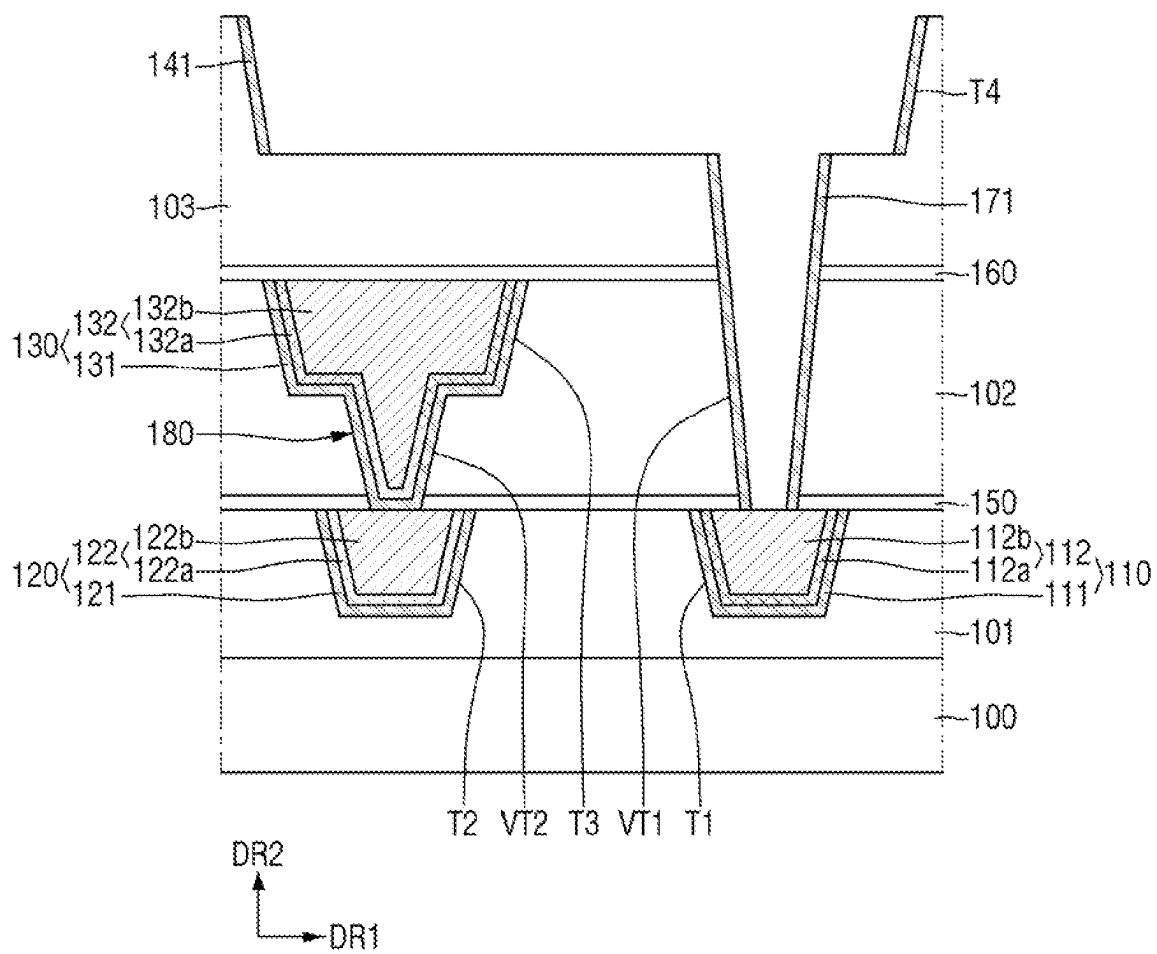


FIG. 10

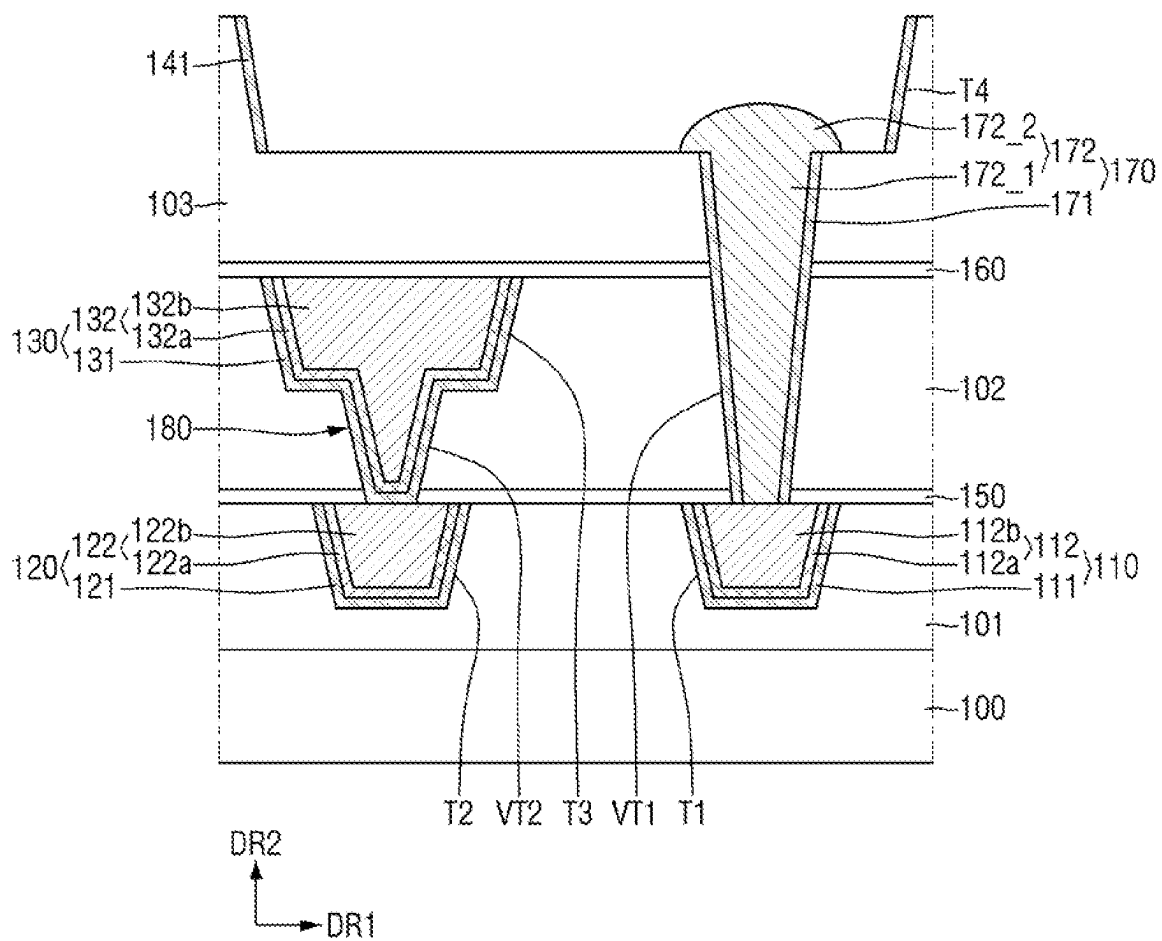


FIG. 12

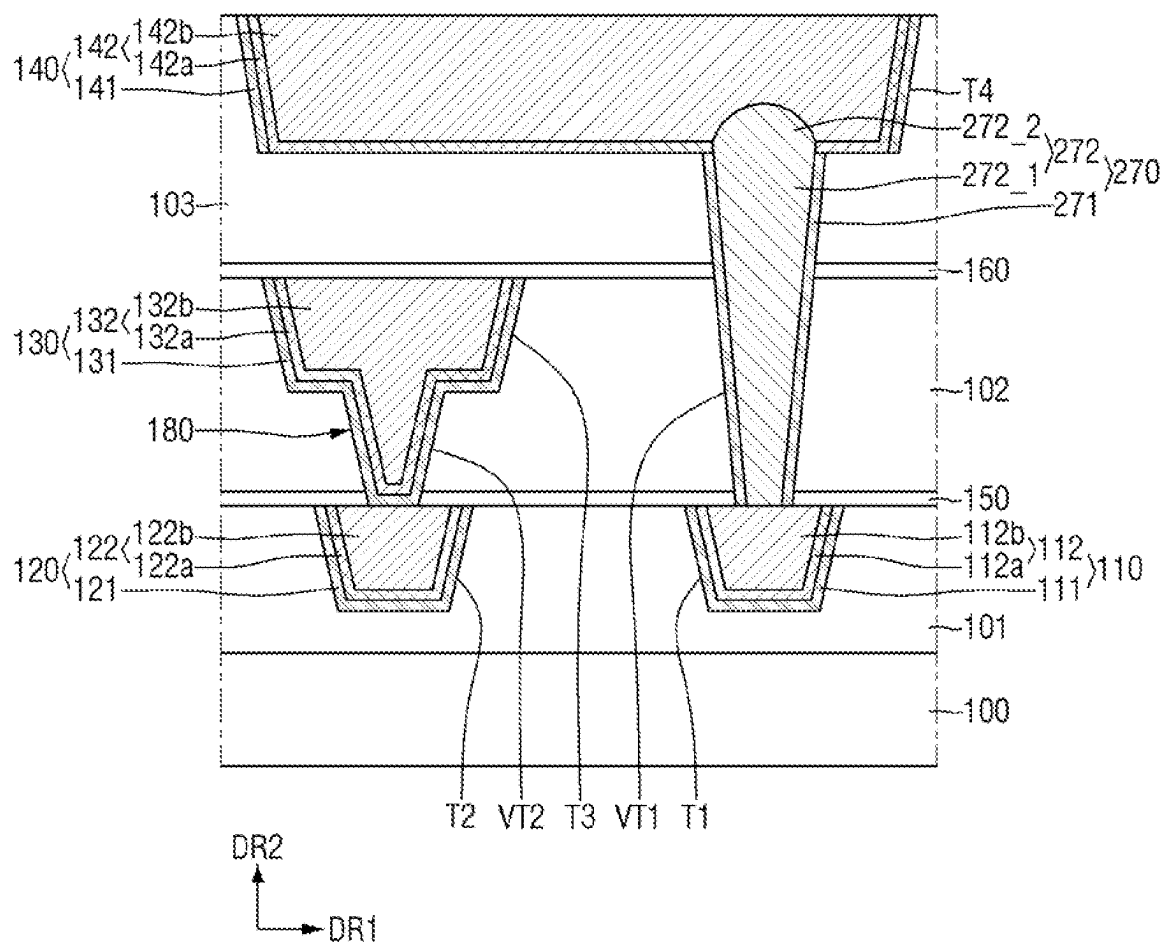


FIG. 13

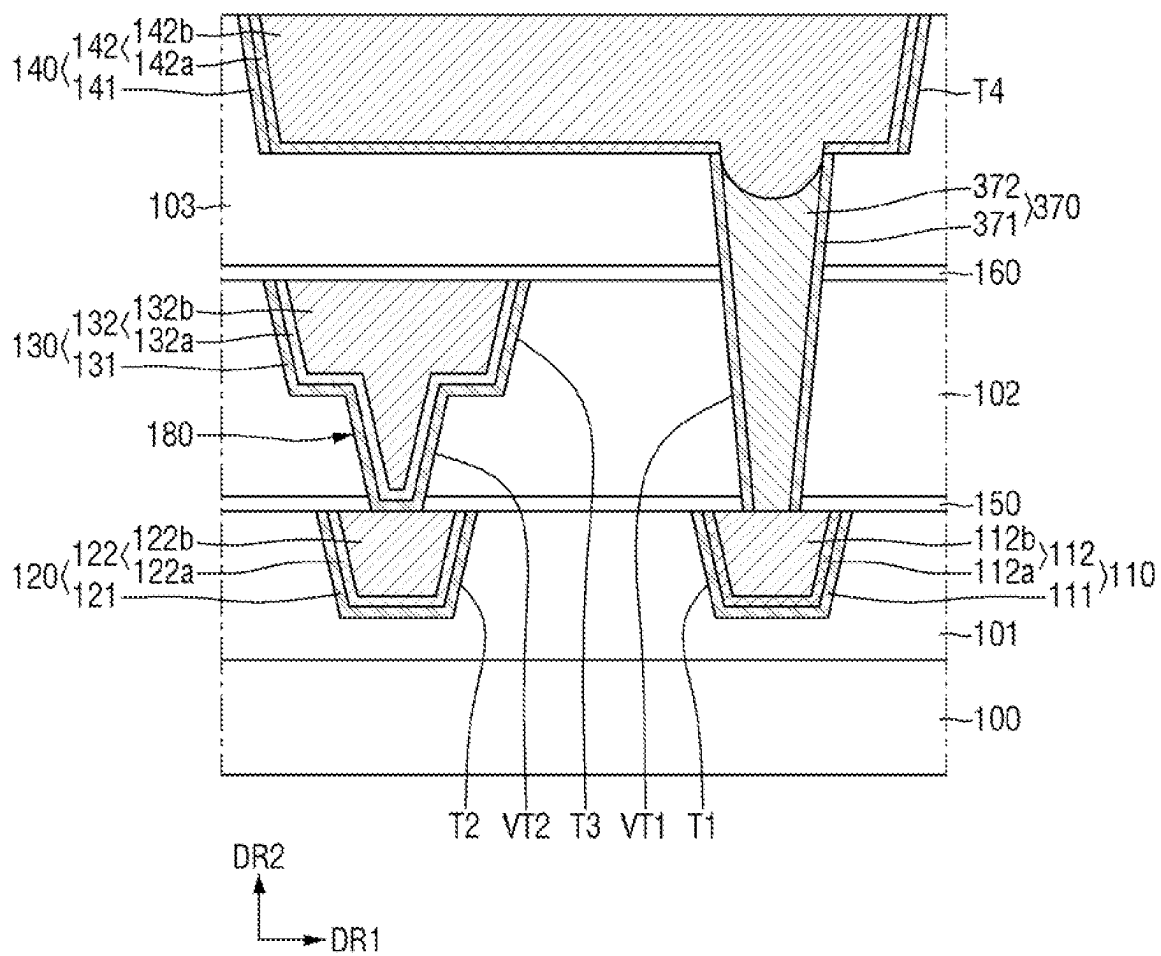


FIG. 14

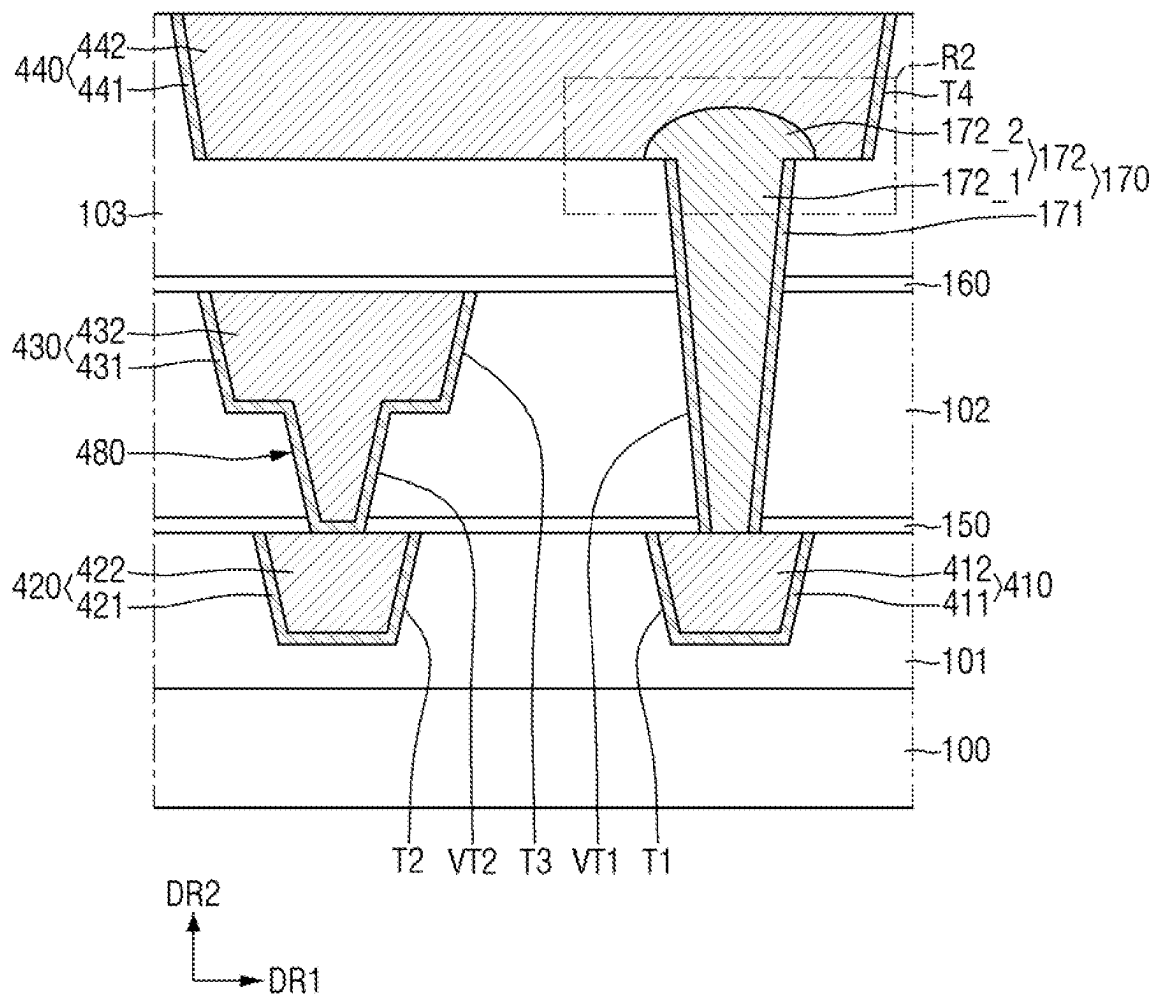


FIG. 15

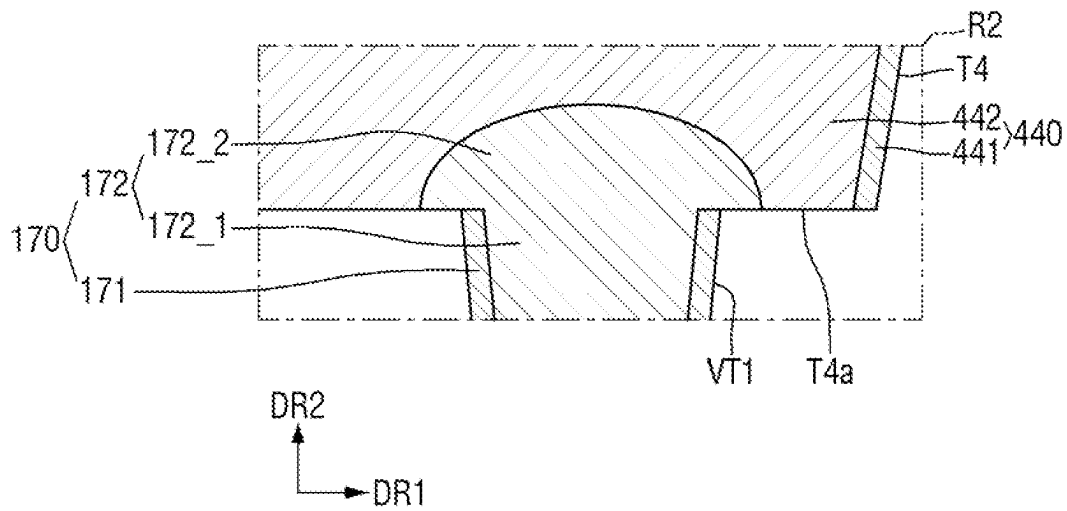


FIG. 18

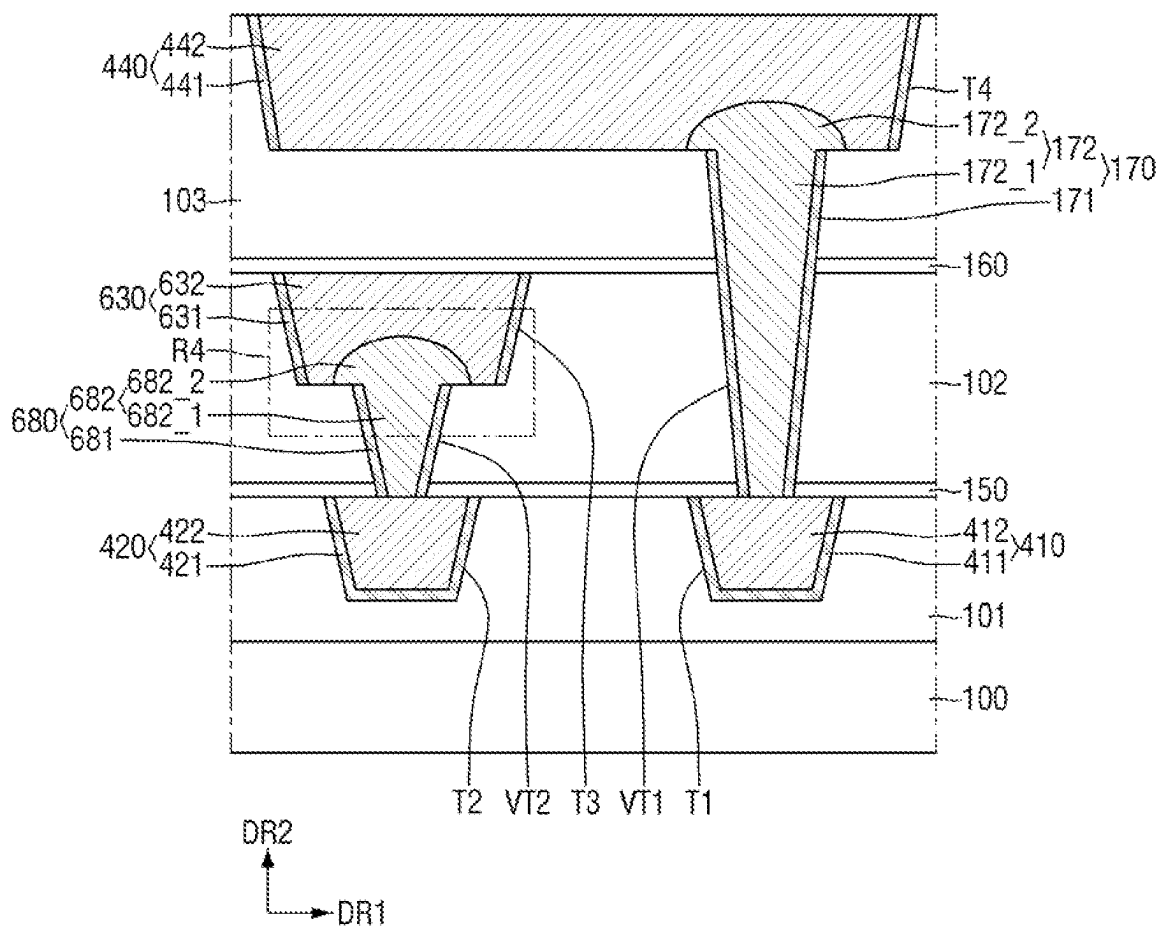
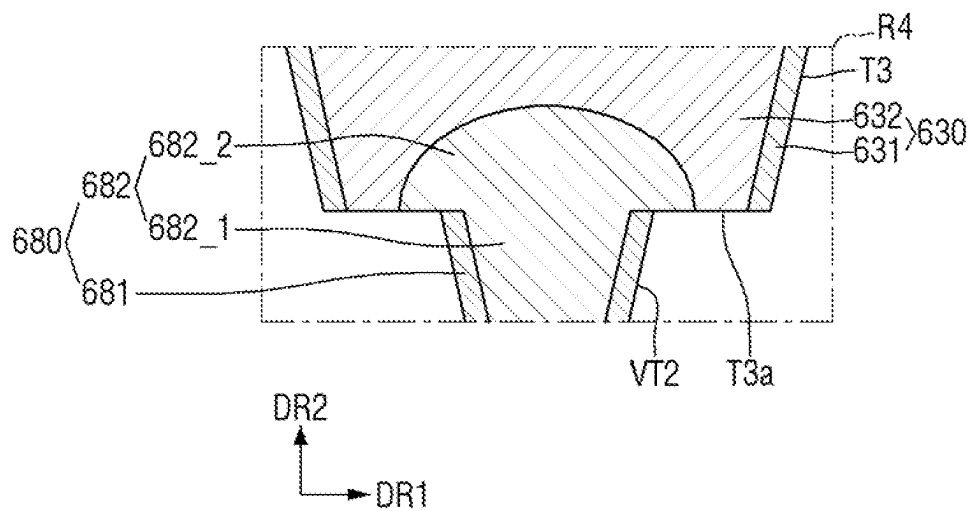


FIG. 19



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SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE SAME

CROSS REFERENCE TO RELATED APPLICATION

This application claims priority under 35 U.S.C. § 119 to Korean Patent Applications No. 10-2021-0095436 filed on Jul. 21, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present inventive concept relates to a semiconductor device and a method for fabricating the same.

DISCUSSION OF THE RELATED ART

As down-scaling of semiconductor elements has progressed rapidly in recent years due to development of electronic technology, high integration and low power consumption of semiconductor chips have been desired. With increased integration in semiconductor chips, gaps between circuit components, such as wirings, have gradually decreased, which may result in increased resistance between circuit components, such as the wiring and a via.

SUMMARY

According to an exemplary embodiment of the present inventive concept, a semiconductor device includes: a substrate; a first interlayer insulating layer disposed on the substrate; a first wiring pattern disposed in a first trench formed in the first interlayer insulating layer; a second interlayer insulating layer disposed on the first interlayer insulating layer; a second wiring pattern disposed in a second trench formed in the second interlayer insulating layer; a third interlayer insulating layer disposed on the second interlayer insulating layer; a third wiring pattern disposed in a third trench formed in the third interlayer insulating layer, and including a first wiring barrier layer and a first wiring filling layer, wherein the first wiring barrier layer is disposed along side walls of the third trench, wherein the first wiring filling layer fills the third trench and is disposed on the first wiring barrier layer, wherein a bottom surface of the first wiring filling layer is in contact with the third interlayer insulating layer; a first via trench extending from an upper surface of the first wiring pattern to a bottom surface of the third trench in a first direction; and a first via including a first via barrier layer and a first via filling layer, wherein the first via barrier layer is disposed along side walls of the first via trench, wherein the first via filling layer fills the first via trench and is disposed on the first via barrier layer, wherein the first via filling layer is in contact with each of the first wiring pattern and the first wiring filling layer, wherein the first wiring filling layer and the first via filling layer include different materials from each other.

According to an exemplary embodiment of the present inventive concept, a semiconductor device includes: a substrate; a first interlayer insulating layer disposed on the substrate; a first wiring pattern disposed in a first trench formed in the first interlayer insulating layer; a first etching stop layer disposed on the first interlayer insulating layer; a second interlayer insulating layer disposed on the first etching stop layer; a second etching stop layer disposed on the second interlayer insulating layer; a third interlayer

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insulating layer disposed on the second etching stop layer; a second wiring pattern disposed in a second trench formed in the third interlayer insulating layer, and including a wiring barrier layer, a liner layer, and a filling layer, wherein the wiring barrier layer is disposed along side walls of the second trench, wherein the liner layer is disposed along side walls of the wiring barrier layer and a bottom surface of the second trench, wherein the filling layer fills the second trench and is disposed on the liner layer, and wherein the liner layer is in contact with the third interlayer insulating layer; a via trench extending from an upper surface of the first wiring pattern to the bottom surface of the second trench in a vertical direction; and a via including a via barrier layer and a via filling layer, wherein the via barrier layer is disposed along side walls of the via trench, wherein the via filling layer fills the via trench and is disposed on the via barrier layer, and wherein the via filling layer is in contact with each of the first wiring pattern and the filling layer, wherein the filling layer and the via filling layer include different materials from each other.

According to an exemplary embodiment of the present inventive concept, a method for fabricating a semiconductor device includes: forming a first interlayer insulating layer on a substrate; forming a first wiring pattern in a first trench formed in the first interlayer insulating layer; forming a second interlayer insulating layer on the first interlayer insulating layer; forming a second wiring pattern in a second trench formed in the second interlayer insulating layer; forming a third interlayer insulating layer on the second interlayer insulating layer; forming a third trench in the third interlayer insulating layer; forming a via trench extending from a bottom surface of the third trench to an upper surface of the first wiring pattern; forming a wiring barrier layer along side walls and the bottom surface of the third trench; forming a via barrier layer along side walls and a bottom surface of the via trench; removing the wiring barrier layer formed on the bottom surface of the third trench and the via barrier layer formed on the bottom surface of the via trench; forming a via filling layer on the via barrier layer and filling the via trench; forming a liner layer along side walls of the wiring barrier layer and the bottom surface of the third trench; and forming a filling layer on the liner layer and the via filling layer to fill the third trench, wherein the filling layer is in contact with the via filling layer.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other features of the present inventive concept will become more apparent by describing in detail exemplary embodiments thereof, with reference to the attached drawings, in which:

FIG. 1 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept;

FIG. 2 is an enlarged view of a region R1 of FIG. 1;

FIGS. 3, 4, 5, 6, 7, 8, 9, 10 and 11 are intermediate stage diagrams illustrating a method for fabricating a semiconductor device according to an exemplary embodiment of the present inventive concept;

FIG. 12 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept;

FIG. 13 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept;

FIG. 14 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept;

FIG. 15 is an enlarged view of a region R2 of FIG. 16;

FIG. 16 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive;

FIG. 17 is an enlarged view of a region R2 of FIG. 14;

FIG. 18 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept; and

FIG. 19 is an enlarged view of a region R4 of FIG. 18.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Hereinafter, a semiconductor device according to an exemplary embodiment of the present inventive concept will be described with reference to FIGS. 1 and 2. In the drawings, like reference numerals may refer to like elements.

FIG. 1 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept. FIG. 2 is an enlarged view of a region R1 of FIG. 1.

Referring to FIGS. 1 and 2, the semiconductor device according to an exemplary embodiment of the present inventive concept includes a substrate 100, a first interlayer insulating layer 101, a second interlayer insulating layer 102, a third interlayer insulating layer 103, a first wiring pattern 110, a second wiring pattern 120, a third wiring pattern 130, a fourth wiring pattern 140, a first etching stop layer 150, a second etching stop layer 160, a first via 170, and a second via 180.

Although the substrate 100 may have a structure in which a base substrate and an epitaxial layer are stacked, the present inventive concept is not limited thereto. For example, the substrate 100 may be a silicon substrate, a gallium arsenide substrate, a silicon germanium substrate, a ceramic substrate, a quartz substrate, a display glass substrate, or the like, or may be an SOI (Semiconductor On Insulator) substrate.

In addition, although not shown, the substrate 100 may include a conductive pattern. Although the conductive pattern may be a metal wiring, a contact, or the like, and may be a gate electrode of a transistor, a source/drain of a transistor, a diode, or the like, the present inventive concept is not limited to.

The first interlayer insulating layer 101 may be disposed on the substrate 100. The first interlayer insulating layer 101 may include, for example, at least one of silicon oxide, silicon nitride, silicon oxynitride and a low dielectric constant material. The first interlayer insulating layer 101 may include a low dielectric constant material to reduce a coupling shape between the wirings.

A first trench T1 may be formed inside the first interlayer insulating layer 101. For example, the first trench T1 may extend from the upper surface of the first interlayer insulating layer 101 toward the substrate 100 in a vertical direction DR2. For example, a bottom surface of the first trench T1 may be formed inside the first interlayer insulating layer 101. However, the present inventive concept is not limited thereto. For example, the bottom surface of the first trench T1 may be coplanar with the bottom surface of the first interlayer insulating layer 101.

Side walls of the first wiring pattern 110 may be surrounded by the first interlayer insulating layer 101. The first

wiring pattern 110 may be disposed inside the first trench T1. The first wiring pattern 110 may include a first wiring barrier layer 111 and a first wiring filling layer 112.

The first wiring barrier layer 111 may be disposed along the side walls and the bottom surface of the first trench T1. The first wiring barrier layer 111 may be formed, for example, conformally. The first wiring barrier layer 111 may include, for example, one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₃), tantalum carbonitride (TaCN), tungsten (W), tungsten nitride (WN), tungsten carbonitride (WCN), zirconium (Zr), zirconium nitride (ZrN), vanadium (V), vanadium nitride (VN), niobium (Nb), niobium nitride (NbN), and combinations thereof. However, the present inventive concept is not limited thereto.

The first wiring filling layer 112 may fill the first trench T1 and may be disposed on the first wiring barrier layer 111. For example, the first wiring filling layer 112 may include a plurality of layers. For example, the first wiring filling layer 112 may include a first liner layer 112a and a first filling layer 112b.

The first liner layer 112a may be disposed on the first wiring barrier layer 111 inside the first trench T1. For example, the first liner layer 112a may be conformally disposed on the first wiring barrier layer 111. The first liner layer 112a may include, for example, at least one of cobalt (Co), ruthenium (Ru) and/or tantalum (Ta). However, the present inventive concept is not limited thereto.

The first filling layer 112b may fill the first trench T1 and may be disposed on the first liner layer 112a. The first filling layer 112b may include, for example, at least one of copper (Cu), carbon (C), silver (Ag), cobalt (Co), tantalum (Ta), indium (In), tin (Sn), zinc (Zn), manganese (Mn), titanium (Ti), magnesium (Mg), chromium (Cr), germanium (Ge), strontium (Sr), platinum (Pt), magnesium (Mg), aluminum (Al), zirconium (Zr), tungsten (W), ruthenium (Ru), iridium (Ir), and/or rhodium (Rh). However, the present inventive concept is not limited thereto.

The second trench T2 may be formed inside the first interlayer insulating layer 101. For example, the second trench T2 may extend from the upper surface of the first interlayer insulating layer 101 toward the substrate 100 in the vertical direction DR2. For example, the bottom surface of the second trench T2 may be formed inside the first interlayer insulating layer 101. The second trench T2 may be spaced apart from the first trench T1 in a horizontal direction DR1. However, the present inventive concept is not limited thereto. For example, the bottom surface of the second trench T2 may be coplanar with the bottom surface of the first interlayer insulating layer 101.

The side walls of the second wiring pattern 120 may be surrounded by the first interlayer insulating layer 101. The second wiring pattern 120 may be disposed inside the second trench T2. The second wiring pattern 120 may be spaced apart from the first wiring pattern 110 in the horizontal direction DR1. For example, the first interlayer insulating layer 101 is between the first wiring pattern 110 and the second wiring pattern 120. The second wiring pattern 120 may include a second wiring barrier layer 121 and a second wiring filling layer 122.

The second wiring barrier layer 121 may be disposed along the side walls and bottom surface of the second trench T2. The second wiring barrier layer 121 may be formed, for example, conformally. For example, the second wiring barrier layer 121 may include the same material as the first wiring barrier layer 111.

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The second wiring filling layer **122** may fill the second trench **T2** and may be disposed on the second wiring barrier layer **121**. For example, the second wiring filling layer **122** may include a plurality of layers. For example, the second wiring filling layer **122** may include a second liner layer **122a** and a second filling layer **122b**.

The second liner layer **122a** may be disposed on the second wiring barrier layer **121** inside the second trench **T2**. For example, the second liner layer **122a** may be conformally disposed on the second wiring barrier layer **121**. For example, the second liner layer **122a** may include the same material as the first liner layer **112a**. The second filling layer **122b** may fill the second trench **T2** and may be disposed on the second liner layer **122a**. For example, the second filling layer **122b** may include the same material as the first filling layer **112b**.

The first etching stop layer **150** may be disposed on the first interlayer insulating layer **101**. The first etching stop layer **150** may cover a part of the upper surfaces of each of the first wiring pattern **110** and the second wiring pattern **120**. Although FIG. 1 shows that the first etching stop layer **150** is formed of a single film, the present inventive concept is not limited thereto. In an exemplary embodiment of the present inventive concept, the first etching stop layer **150** may include a plurality of layers.

The first etching stop layer **150** may include, for example, silicon nitride, silicon oxynitride, silicon carbon nitride, and the like. In an exemplary embodiment of the present inventive concept, the first etching stop layer **150** may include a metal, for example, aluminum (Al). However, the present inventive concept is not limited thereto.

The second interlayer insulating layer **102** may be disposed on the first etching stop layer **150**. The second interlayer insulating layer **102** may include, for example, at least one of silicon oxide, silicon nitride, silicon oxynitride and/or a low dielectric constant material. The second interlayer insulating layer **102** may include a low dielectric constant material to reduce the coupling shape between the wirings (e.g., **130** and **140**).

The third trench **T3** may be formed inside the second interlayer insulating layer **102**. For example, the third trench **T3** may extend from the upper surface of the second interlayer insulating layer **102** toward the substrate **100** in the vertical direction **DR2**. For example, the bottom surface of the third trench **T3** may be formed inside the second interlayer insulating layer **102**.

The second via trench **VT2** may extend from the upper surface of the second wiring pattern **120** to the bottom surface of the third trench **T3** in the vertical direction **DR2**. A width of the upper surface of the second via trench **VT2** in the horizontal direction **DR1** may be smaller than a width of the bottom surface of the third trench **T3** in the horizontal direction **DR1**. Further, a width of the bottom surface of the second via trench **VT2** in the horizontal direction **DR1** may be smaller than the width of the upper surface of the second trench **T2** in the horizontal direction **DR1**.

The side walls of the third wiring pattern **130** may be surrounded by the second interlayer insulating layer **102**. The third wiring pattern **130** may be disposed inside the third trench **T3**. The side walls of the second via **180** may be surrounded by each of the second interlayer insulating layer **102** and the first etching stop layer **150**. The second via **180** may be disposed inside the second via trench **VT2**. Each of the third wiring pattern **130** and the second via **180** may include a third wiring barrier layer **131** and a third wiring filling layer **132**.

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The third wiring barrier layer **131** may be disposed along the side walls and bottom surfaces of each of the third trench **T3** and the second via trench **VT2**. The third wiring barrier layer **131** may be formed, for example, conformally in the third trench **T3** and the second via trench **VT2**. For example, the third wiring barrier layer **131** may include the same material as the first wiring barrier layer **111**.

The third wiring filling layer **132** may fill each of the third trench **T3** and the second via trench **VT2**, and may be disposed on the third wiring barrier layer **131**. For example, the third wiring filling layer **132** may include a plurality of layers. For example, the third wiring filling layer **132** may include a third liner layer **132a** and a third filling layer **132b**.

The third liner layer **132a** may be disposed on the third wiring barrier layer **131** inside each of the third trench **T3** and the second via trench **VT2**. For example, the third liner layer **132a** may be conformally disposed on the third wiring barrier layer **131**. For example, the third liner layer **132a** may include the same material as the first liner layer **112a**. The third filling layer **132b** may fill each of the third trench **T3** and the second via trench **VT2**, and may be disposed on the third liner layer **132a**. For example, the third filling layer **132b** may include the same material as the first filling layer **112b**.

The second etching stop layer **160** may be disposed on the second interlayer insulating layer **102**. For example, the second etching stop layer **160** may cover at least a part of the upper surface of the third wiring pattern **130**. Although FIG. 1 shows that the second etching stop layer **160** is formed of a single film, the present inventive concept is not limited thereto. In an exemplary embodiment of the present inventive concept, the second etching stop layer **160** may include a plurality of layers.

The second etching stop layer **160** may include, for example, silicon nitride, silicon oxynitride, silicon carbon nitride, and the like. In an exemplary embodiment of the present inventive concept, the second etching stop layer **160** may include a metal, for example, aluminum (Al). However, the present inventive concept is not limited thereto.

The third interlayer insulating layer **103** may be disposed on the second etching stop layer **160**. The third interlayer insulating layer **103** may include, for example, at least one of silicon oxide, silicon nitride, silicon oxynitride and/or a low dielectric constant material. The third interlayer insulating layer **103** may include a low dielectric constant material to reduce the coupling shape between the wirings.

A fourth trench **T4** may be formed inside the third interlayer insulating layer **103**. For example, the fourth trench **T4** may extend from the upper surface of the third interlayer insulating layer **103** toward the substrate **100** in the vertical direction **DR2**. For example, a bottom surface **T4a** of the fourth trench **T4** may be formed inside the third interlayer insulating layer **103**.

The side walls of the fourth wiring pattern **140** may be surrounded by the third interlayer insulating layer **103**. The fourth wiring pattern **140** may be disposed inside the fourth trench **T4**. The fourth wiring pattern **140** may include a fourth wiring barrier layer **141** and a fourth wiring filling layer **142**.

The fourth wiring barrier layer **141** may be disposed along the side walls of the fourth trench **T4**. For example, the fourth wiring barrier layer **141** is not disposed on the bottom surface of the fourth trench **T4**. However, the present inventive concept is not limited thereto. For example, the fourth wiring barrier layer **141** may be disposed on a portion of the bottom surface of the fourth trench **T4**. The fourth wiring barrier layer **141** may be formed, for example, conformally.

The fourth wiring barrier layer **141** may include, for example, one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), tantalum carbonitride (TaCN), tungsten (W), tungsten nitride (WN), tungsten carbonitride (WCN), zirconium (Zr), zirconium nitride (ZrN), vanadium (V), vanadium nitride (VN), niobium (Nb), niobium nitride (NbN), and combinations thereof. However, the present inventive concept is not limited thereto.

The fourth wiring filling layer **142** may fill the fourth trench **T4** and may be disposed on the fourth wiring barrier layer **141**. For example, the fourth wiring filling layer **142** may include a plurality of layers. For example, the fourth wiring filling layer **142** may include a fourth liner layer **142a** and a fourth filling layer **142b**.

A bottom surface of the fourth wiring filling layer **142** may be in contact with the third interlayer insulating layer **103**. For example, a part of the bottom surface of the fourth wiring filling layer **142** may be in contact with the third interlayer insulating layer **103** on both sides of the first via **170**.

The fourth liner layer **142a** may be disposed inside the fourth trench **T4** along the side walls of the fourth wiring barrier layer **141** and the bottom surface of the fourth trench **T4**. For example, the fourth liner layer **142a** may be in contact with the third interlayer insulating layer **103** on both sides of the first via **170**. For example, the fourth liner layer **142a** may be conformally disposed in the fourth trench **T4**. The fourth liner layer **142a** may include, for example, at least one of cobalt (Co), ruthenium (Ru) and/or tantalum (Ta). However, the present inventive concept is not limited thereto.

The fourth filling layer **142b** may fill the fourth trench **T4** and may be disposed on the fourth liner layer **142a**. The fourth filling layer **142b** may include, for example, at least one of copper (Cu), carbon (C), silver (Ag), cobalt (Co), tantalum (Ta), indium (In), tin (Sn), zinc (Zn), manganese (Mn), titanium (Ti), magnesium (Mg), chromium (Cr), germanium (Ge), strontium (Sr), platinum (Pt), magnesium (Mg), aluminum (Al), zirconium (Zr), tungsten (W), ruthenium (Ru), iridium (Ir), and/or rhodium (Rh). However, the present disclosure is not limited thereto.

The first via trench **VT1** may extend from the upper surface of the first wiring pattern **110** to the bottom surface of the fourth trench **T4** in the vertical direction **DR2**. The first via trench **VT1** may penetrate the first etching stop layer **150**, the second interlayer insulating layer **102**, the second etching stop layer **160**, and a part of the third interlayer insulating layer **103** in the vertical direction **DR2**.

A width of the upper surface of the first via trench **VT1** in the horizontal direction **DR1** may be smaller than a width of the bottom surface **T4a** of the fourth trench **T4** in the horizontal direction **DR1**. Further, the width of the bottom surface of the first via trench **VT1** in the horizontal direction **DR1** may be smaller than the width of the upper surface of the first trench **T1** in the horizontal direction **DR1**.

The side walls of the first via **170** may be surrounded by each of the first etching stop layer **150**, the second interlayer insulating layer **102**, the second etching stop layer **160**, and the third interlayer insulating layer **103**. The first via **170** may be disposed inside the first via trench **VT1**. A part of the first via **170** may protrude into the fourth trench **T4**. For example, the upper surface of a part of the first via **170** protruding into the fourth trench **T4** may be formed to be higher than the bottom surface **T4a** of the fourth trench **T4**. For example, the upper surface of the part of the first via **170** protruding into the fourth trench **T4** may protrude beyond

the bottom surface **T4a** of the fourth trench. The first via **170** may be spaced apart from the third wiring pattern **130** in the horizontal direction **DR1**.

The first via **170** may include a first via barrier layer **171** and a first via filling layer **172**. The first via barrier layer **171** may be disposed along the side walls of the first via trench **VT1**. For example, the first via barrier layer **171** may be conformally disposed in the first via trench **VT1**. For example, the first via barrier layer **171** may include the same material as the fourth wiring barrier layer **141**. The first via barrier layer **171** may include, for example, at least one of cobalt (Co), ruthenium (Ru) and/or tantalum (Ta). However, the present inventive concept is not limited thereto.

The first via filling layer **172** may fill the first via trench **VT1** and may be disposed on the first via barrier layer **171**. A part of the first via filling layer **172** may be disposed inside the fourth trench **T4**. The first via filling layer **172** may include a first portion **172_1** disposed inside the first via trench **VT1**, and a second portion **172_2** disposed inside the fourth trench **T4**. The second portion **172_2** of the first via filling layer **172** may be a portion protruding into the fourth trench **T4**. The bottom surface of the first portion **172_1** of the first via filling layer **172** may be in contact with the first wiring pattern **110**. For example, the bottom surface of the first portion **172_1** of the first via filling layer **172** may be in direct contact with the first wiring pattern **110**. The upper surface of the second portion **172_2** of the first via filling layer **172** may be in contact with the fourth filling layer **142b**. For example, the upper surface of the second portion **172_2** of the first via filling layer **172** may be in direct contact with the fourth filling layer **142b**.

At least a part of the second portion **172_2** of the first via filling layer **172** may protrude beyond the first portion **172_1** of the first via filling layer **172** in the horizontal direction **DR1**. For example, the part of the second portion **172_2** of the first via filling layer **172** may protrude beyond side surfaces of the first portion **172_1** of the first via filling layer **172** in the horizontal direction **DR1**. A bottom surface of a part of the second portion **172_2** of the first via filling layer **172** protruding beyond the first portion **172_1** of the first via filling layer **172** in the horizontal direction **DR1** may be in contact with each of an uppermost surface of the first via barrier layer **171** and the third interlayer insulating layer **103**.

A width of the bottom surface of the second portion **172_2** of the first via filling layer **172** in the horizontal direction **DR1** may be greater than a width of the upper surface (or, e.g., an upper portion) of the first portion **172_1** of the first via filling layer **172** in the horizontal direction **DR1**.

The first via filling layer **172** may include, for example, at least one of copper (Cu), carbon (C), silver (Ag), cobalt (Co), tantalum (Ta), indium (In), tin (Sn), zinc (Zn), manganese (Mn), titanium (Ti), magnesium (Mg), chromium (Cr), germanium (Ge), strontium (Sr), platinum (Pt), magnesium (Mg), aluminum (Al), zirconium (Zr), tungsten (W), ruthenium (Ru), iridium (Ir), and/or rhodium (Rh). However, the present inventive concept is not limited thereto. The first via filling layer **172** may include a material different from the fourth filling layer **142b**. For example, the fourth filling layer **142b** may include copper (Cu), and the first via filling layer **172** may include tungsten (W).

In the semiconductor device according to an exemplary embodiment of the present inventive concept, because the first via filling layer **172** included in the first via **170** is formed to be in contact with the first (e.g., lower) wiring pattern **110**, the resistance between the first via **170** and the first wiring pattern **110** can be reduced. Further, in the

semiconductor device according to an exemplary embodiment of the present inventive concept, because the first via filling layer **172** included in the first via **170** is formed to be in contact with the fourth filling layer **142b** included in the fourth (e.g., upper) wiring pattern **140**, the resistance between the first via **170** and the fourth wiring pattern **140** can be reduced.

Hereinafter, a method for fabricating a semiconductor device according to an exemplary embodiment of the present inventive concept will be described with reference to FIGS. 1, 3 to 11.

FIGS. 3 to 11 are intermediate stage diagrams illustrating a method for fabricating a semiconductor device according to an exemplary embodiment of the present inventive concept.

Referring to FIG. 3, the first interlayer insulating layer **101** may be formed on the substrate **100**. Subsequently, a first trench **T1** and a second trench **T2** may be formed inside the first interlayer insulating layer **101** to be spaced apart from each other in the horizontal direction **DR1**. Subsequently, the first wiring pattern **110** may be formed inside the first trench **T1**, and the second wiring pattern **120** may be formed inside the second trench **T2**.

The first wiring pattern **110** may include a first wiring barrier layer **111**, a first liner layer **112a**, and a first filling layer **112b**. The first wiring barrier layer **111** may be formed along the side walls and the bottom surface of the first trench **T1**, and the first liner layer **112a** may be formed on the first wiring barrier layer **111**. The first filling layer **112b** may be formed to fill the first trench **T1** and may be disposed on the first liner layer **112a**.

The second wiring pattern **120** may include a second wiring barrier layer **121**, a second liner layer **122a**, and a second filling layer **122b**. The second wiring barrier layer **121** may be formed along the side walls and the bottom surface of the second trench **T2**, and the second liner layer **122a** may be formed on the second wiring barrier layer **121**. The second filling layer **122b** may be formed to fill the second trench **T2** and may be disposed on the second liner layer **122a**.

Subsequently, the first etching stop layer **150** and the second interlayer insulating layer **102** may be formed sequentially on the first interlayer insulating layer **101**, the upper surface of the first wiring pattern **110** and the upper surface of the second wiring pattern **120**.

Referring to FIG. 4, a third trench **T3** may be formed inside the second interlayer insulating layer **102** and may penetrate the second interlayer insulating layer **102**. Subsequently, a second via trench **VT2** extending from the bottom surface of the third trench **T3** to the upper surface of the second wiring pattern **120** may be formed. A width of the upper surface of the second via trench **VT2** in the horizontal direction **DR1** may be formed to be smaller than a width of the bottom surface of the third trench **T3** in the horizontal direction **DR1**.

Referring to FIG. 5, a third wiring pattern **130** may be formed inside the third trench **T3**. Further, a second via **180** may be formed inside the second via trench **VT2**. The third wiring pattern **130** and the second via **180** may be formed through the same fabricating process. However, the present inventive concept is not limited thereto. For example, the third wiring pattern **130** and the second via **180** may be formed through separate fabricating processes.

The third wiring pattern **130** and the second via **180** may include a third wiring barrier layer **131**, a third liner layer **132a**, and a third filling layer **132b**. The third wiring barrier layer **131** may be formed along the side walls and the bottom

surface of each of the third trench **T3** and the second via trench **VT2**, and the third liner layer **132a** may be formed on the third wiring barrier layer **131**. The third filling layer **132b** may be formed to fill each of the third trench **T3** and the second via trench **VT2** and may be disposed on the third liner layer **132a**.

Referring to FIG. 6, the second etching stop layer **160** and the third interlayer insulating layer **103** may be sequentially formed on the upper surfaces of the second interlayer insulating layer **102** and the third wiring pattern **130**.

Referring to FIG. 7, a fourth trench **T4** may be formed inside the third interlayer insulating layer **103**. Subsequently, a first via trench **VT1** extending from the bottom surface of the fourth trench **T4** to the upper surface of the first wiring pattern **110** may be formed. The first via trench **VT1** may be formed to penetrate the first etching stop layer **150**, the second interlayer insulating layer **102**, the second etching stop layer **160**, and a part of the third interlayer insulating layer **103** in the vertical direction **DR2**. A width of the upper surface of the first via trench **VT1** in the horizontal direction **DR1** may be smaller than the width of the bottom surface of the fourth trench **T4** in the horizontal direction **DR1**.

Referring to FIG. 8, a fourth wiring barrier layer **141** may be formed along the side walls and bottom surface of the fourth trench **T4**. Further, a first via barrier layer **171** may be formed along the side walls and bottom surface of the first via trench **VT1**. For example, the fourth wiring barrier layer **141** may contact the first wiring pattern **110**. The fourth wiring barrier layer **141** and the first via barrier layer **171** may include the same material. The fourth wiring barrier layer **141** and the first via barrier layer **171** may be formed through the same fabricating process.

Referring to FIG. 9, the fourth wiring barrier layer **141** formed on the bottom surface of the fourth trench **T4** may be removed. For example, at least a portion of the fourth wiring barrier layer **141** formed on the bottom surface of the fourth trench **T4** may be removed. Accordingly, the third interlayer insulating layer **103** may be exposed on the bottom surface of the fourth trench **T4**. Further, the first via barrier layer **171** formed on the bottom surface of the first via trench **VT1** may be removed. For example, at least a portion of the first via barrier layer **171** formed on the bottom surface of the fourth trench **T4** may be removed. As a result, the first wiring pattern **110** may be exposed on the bottom surface of the first via trench **VT1**.

The fourth wiring barrier layer **141** formed on the bottom surface of the fourth trench **T4** and the first via barrier layer **171** formed on the bottom surface of the first via trench **VT1** may be removed through the same fabricating process.

Referring to FIG. 10, the first via filling layer **172** may be formed to fill the inside of the first via trench **VT1** on the first via barrier layer **171**. The first portion **172_1** of the first via filling layer **172** may be formed inside the first via trench **VT1**. The first portion **172_1** of the first via filling layer **172** may be in contact with the first wiring pattern **110**.

Further, the second portion **172_2** of the first via filling layer **172** may be formed to protrude into the fourth trench **T4** on the first portion **172_1** of the first via filling layer **172**. At least a part of the second portion **172_2** of the first via filling layer **172** may extend along the bottom surface of the fourth trench **T4** in the horizontal direction **DR1**. For example, at least a part of the second portion **172_2** of the first via filling layer **172** may be formed on the bottom surface of the fourth trench **T4** on both sides of the first via trench **VT1**. The width of the bottom surface of the second portion **172_2** of the first via filling layer **172** in the

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horizontal direction DR1 may be greater than the width the upper surface of the first via trench VT1 in the horizontal direction DR1.

Referring to FIG. 11, a fourth liner layer 142a may be formed inside the fourth trench T4 along the side walls of the fourth wiring barrier layer 141 and the bottom surface of the fourth trench T4. The fourth liner layer 142a is not formed on the second portion 172_2 of the first via filling layer 172. For example, the second portion 172_2 of the first via filling layer 172 may be exposed by the fourth trench T4. As another example, the second portion 172_2 of the first via filling layer 172 may be exposed by the fourth trench T4 on the fourth liner layer 142a. The fourth liner layer 142a may be in contact with the third interlayer insulating layer 103.

Referring to FIG. 1, a fourth filling layer 142b may be formed to fill the fourth trench T4, and may be disposed on the fourth liner layer 142a and the second portion 172_2 of the first via filling layer 172. The fourth filling layer 142b may be in contact with the second portion 172_2 of the first via filling layer 172. The semiconductor device shown in FIG. 1 may be fabricated through such a fabricating process.

Hereinafter, a semiconductor device according to an exemplary embodiment of the present inventive concept will be described referring to FIG. 12. Differences from the semiconductor device shown in FIGS. 1 and 2 will be mainly described, and thus, redundant descriptions of components may be omitted.

FIG. 12 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept.

Referring to FIG. 12, in the semiconductor device according to an exemplary embodiment of the present inventive concept, a second portion 272_2 of a second via filling layer 272 of a first via 270 may overlap a first portion 272_1 of the second via filling layer 272 of the first via 270 in the vertical direction DR2. For example, the second portion 272_2 of the second via filling layer 272 does not protrude from the side walls of the first portion 272_1 of the second via filling layer 272 in the horizontal direction DR1. An uppermost surface of the first via barrier layer 271 may be in contact with the fourth liner layer 142a.

Hereinafter, a semiconductor device according to an exemplary embodiment of the present inventive concept will be described with reference to FIG. 13. Differences from the semiconductor device shown in FIGS. 1 and 2 will be mainly described, and thus, redundant descriptions of components may be omitted.

FIG. 13 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept.

Referring to FIG. 13, in the semiconductor device according to an exemplary embodiment of the present inventive concept, an upper surface of a first via filling layer 372 of a first via 370 may be formed to be concave toward the substrate 100. At least a part of the fourth wiring filling layer 142 may be disposed inside the first via trench VT1. For example, at least a part of the fourth filling layer 142b may be disposed inside the first via trench VT1. For example, the fourth filling layer 142b may be disposed on the concave surface of the first via filling layer 372.

The fourth filling layer 142b may be in contact with the first via filling layer 372 inside the first via trench VT1. An uppermost surface of the first via barrier layer 371 may be in contact with the fourth liner layer 142a.

Hereinafter, a semiconductor device according to an exemplary embodiment of the present inventive concept will be described with reference to FIGS. 14 and 15. Differences

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from the semiconductor device shown in FIGS. 1 and 2 will be mainly described, and thus, redundant descriptions of components may be omitted.

FIG. 14 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept. FIG. 15 is an enlarged view of a region R2 of FIG. 14.

Referring to FIGS. 14 and 15, in the semiconductor device according to an exemplary embodiment of the present inventive concept, a first wiring filling layer 412, a second wiring filling layer 422, a third wiring filling layer 432, and a fourth wiring filling layer 442 may each be formed by a single film.

For example, the first wiring pattern 410 may include a first wiring barrier layer 411 disposed along the side walls and the bottom surface of the first trench T1, and a first wiring filling layer 412 that fills the first trench T1 and is disposed on the first wiring barrier layer 411. The second wiring pattern 420 may include a second wiring barrier layer 421 disposed along the side walls and bottom surface of the second trench T2, and a second wiring filling layer 422 that fills the second trench T2 and is disposed on the second wiring barrier layer 421. The third wiring pattern 430 may include a third wiring barrier layer 431 disposed along the side walls and bottom surface of the third trench T3, and a third wiring filling layer 432 that fills the third trench T3 and is disposed on the third wiring barrier layer 431. The fourth wiring pattern 440 may include a fourth wiring barrier layer 441 disposed along the side walls and on the bottom surface of the fourth trench T4, and a fourth wiring filling layer 442 that fills the fourth trench T4 and is disposed on the fourth wiring barrier layer 441.

A second via 480 may include a third wiring barrier layer 431 disposed along the side walls and bottom surface of the second via trench VT2, and a third wiring filling layer 432 that fills the second via trench VT2 and is disposed on the third wiring barrier layer 431.

Hereinafter, a semiconductor device according to an exemplary embodiment of the present inventive concept will be described with reference to FIGS. 16 and 17. Differences from the semiconductor device shown in FIGS. 1 and 2 will be mainly described, and thus, redundant descriptions of components may be omitted.

FIG. 16 is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept. FIG. 17 is an enlarged view of a region R3 of FIG. 16.

Referring to FIGS. 16 and 17, in the semiconductor device according to an exemplary embodiment of the present inventive concept, a part of the second via 580 may protrude into the third trench T3. For example, an upper surface of a part of the second via 580 protruding into the third trench T3 may be formed to be higher than a bottom surface T3a of the third trench T3.

The third wiring pattern 530 may include a third wiring barrier layer 531 disposed along the side walls of the third trench T3, and a third wiring filling layer 532 that fills the third trench T3 and is disposed on the third wiring barrier layer 531. The third wiring filling layer 532 may include a third liner layer 532a and a third filling layer 532b.

The third liner layer 532a may be disposed along the side walls of the third wiring barrier layer 531 and the bottom surface T3a of the third trench T3. The third filling layer 532b may fill the third trench T3 and may be disposed on the third liner layer 532a. The third liner layer 532a may be in contact with the second interlayer insulating layer 102 on both sides of the second via trench VT2.

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The second via **580** may include a second via barrier layer **581** and a second via filling layer **582**. The second via barrier layer **581** may be disposed along the side walls of the second via trench **VT2**. For example, the second via barrier layer **581** may be conformally disposed in the second via trench **VT2**. For example, the second via barrier layer **581** may include the same material as the third wiring barrier layer **531**. The second via barrier layer **581** may include, for example, at least one of cobalt (Co), ruthenium (Ru) and/or tantalum (Ta). However, the present inventive concept is not limited thereto.

The second via filling layer **582** may fill the second via trench **VT2** and may be disposed on the second via barrier layer **581**. A part of the second via filling layer **582** may be disposed inside the third trench **T3**. The second via filling layer **582** may include a first portion **582_1** disposed inside the second via trench **VT2**, and a second portion **582_2** disposed inside the third trench **T3**. The second portion **582_2** of the second via filling layer **582** may be a portion that protrudes into the third trench **T3**. A bottom surface of the first portion **582_1** of the second via filling layer **582** may be in contact with the second wiring pattern **120**. An upper surface of the second portion **582_2** of the second via filling layer **582** may be in contact with the third filling layer **532b**.

At least a part of the second portion **582_2** of the second via filling layer **582** may protrude in the horizontal direction **DR1** beyond the first portion **582_1** of the second via filling layer **582**. A bottom surface of a part of the second portion **582_2** of the second via filling layer **582** protruding beyond the first portion **582_1** of the second via filling layer **582** in the horizontal direction **DR1** may be in contact with each of the uppermost surface of the second via barrier layer **581** and the second interlayer insulating layer **102**.

A width of the bottom surface of the second portion **582_2** of the second via filling layer **582** in the horizontal direction **DR1** may be greater than a width of the upper surface of the first portion **582_1** of the second via filling layer **582** in the horizontal direction **DR1**.

Hereinafter, a semiconductor device according to an exemplary embodiment of the present inventive concept will be described with reference to FIGS. **18** and **19**. Differences from the semiconductor device shown in FIGS. **14** and **15** will be mainly described, and thus, redundant descriptions of components may be omitted.

FIG. **18** is a diagram illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept. FIG. **19** is an enlarged view of a region **R4** of FIG. **18**.

Referring to FIGS. **18** and **19**, in the semiconductor device according to an exemplary embodiment of the present inventive concept, a part of a second via **680** may protrude into the third trench **T3**. For example, an upper surface of a part of the second via **680** protruding into the third trench **T3** may be formed to be higher than the bottom surface **T3a** of the third trench **T3**.

A third wiring pattern **630** may include a third wiring barrier layer **631** disposed along the side walls of the third trench **T3**, and a third wiring filling layer **632** that fills the third trench **T3** and may be disposed on the third wiring barrier layer **631**. The third wiring filling layer **632** may be in contact with the second interlayer insulating layer **102** on both sides of the second via trench **VT2**.

The second via **680** may include a second via barrier layer **681** and a second via filling layer **682**. The second via barrier layer **681** may be disposed along the side walls of the second via trench **VT2**. For example, the second via barrier layer

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681 may be conformally disposed in the second via trench **VT2**. For example, the second via barrier layer **681** may include the same material as the third wiring barrier layer **631**. The second via barrier layer **681** may include, for example, at least one of cobalt (Co), ruthenium (Ru) and/or tantalum (Ta). However, the present inventive concept is not limited thereto.

The second via filling layer **682** may fill the second via trench **VT2** and may be disposed on the second via barrier layer **681**. A part of the second via filling layer **682** may be disposed inside the third trench **T3**. The second via filling layer **682** may include a first portion **682_1** disposed inside the second via trench **VT2**, and a second portion **682_2** disposed inside the third trench **T3**. The second portion **682_2** of the second via filling layer **682** may be a portion that protrudes into the third trench **T3**. A bottom surface of the first portion **682_1** of the second via filling layer **682** may be in contact with the second wiring pattern **420**. An upper surface of the second portion **682_2** of the second via filling layer **682** may be in contact with the third wiring filling layer **632**.

At least a part of the second portion **682_2** of the second via filling layer **682** may protrude in the horizontal direction **DR1** beyond the first portion **682_1** of the second via filling layer **682**. The bottom surface of a part of the second portion **682_2** of the second via filling layer **682** protruding in the horizontal direction **DR1** beyond the first portion **682_1** of the second via filling layer **682** may be in contact with each of the uppermost surface of the second via barrier layer **681** and the second interlayer insulating layer **102**.

A width of the bottom surface of the second portion **682_2** of the second via filling layer **682** in the horizontal direction **DR1** may be greater than a width of the upper surface (or, e.g., an upper portion) of the first portion **682_1** of the second via filling layer **682** in the horizontal width **DR1**.

An exemplary embodiment of the present inventive concept provides a semiconductor device in which resistance between a via and a lower wiring pattern may be reduced by forming a via filling layer, which is included in the via, to be in contact with the lower wiring pattern.

In addition, an exemplary embodiment of the present inventive concept provides a semiconductor device in which the resistance between a via and an upper wiring pattern may be reduced by forming a via filling layer, which is included in the via, to be in contact with a filling layer included in the upper wiring pattern.

While the present inventive concept has been described with reference to exemplary embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made thereto without departing from the spirit and scope of the present invention.

What is claimed is:

1. A semiconductor device comprising:

- a substrate;
- a first interlayer insulating layer disposed on the substrate;
- a first wiring pattern disposed in a first trench formed in the first interlayer insulating layer;
- a second interlayer insulating layer disposed on the first interlayer insulating layer;
- a second wiring pattern disposed in a second trench formed in the second interlayer insulating layer;
- a third interlayer insulating layer disposed on the second interlayer insulating layer;
- a third wiring pattern disposed in a third trench formed in the third interlayer insulating layer, and including a first wiring barrier layer and a first wiring filling layer, wherein the first wiring barrier layer is disposed along

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side walls of the third trench, wherein the first wiring filling layer fills the third trench and is disposed on the first wiring barrier layer, wherein a bottom surface of the first wiring filling layer is in contact with the third interlayer insulating layer;

a first via trench extending from an upper surface of the first wiring pattern to a bottom surface of the third trench in a first direction; and

a first via including a first via barrier layer and a first via filling layer, wherein the first via barrier layer is disposed along side walls of the first via trench, wherein the first via filling layer fills the first via trench and is disposed on the first via barrier layer, wherein the first via filling layer is in contact with each of the first wiring pattern and the first wiring filling layer,

wherein the first wiring filling layer and the first via filling layer include different materials from each other.

2. The semiconductor device of claim 1, wherein the first wiring filling layer includes:

a first liner layer disposed along side walls of the first wiring barrier layer and a bottom surface of the third trench; and

a first filling layer filling the third trench and disposed on the first liner layer and in contact with the first via filling layer.

3. The semiconductor device of claim 1, wherein the first via filling layer includes

a first portion disposed inside the first via trench; and

a second portion protruding into the third trench and connected to the first portion of the first via trench.

4. The semiconductor device of claim 3, wherein at least a part of a bottom surface of the second portion of the first via filling layer is in contact with the third interlayer insulating layer.

5. The semiconductor device of claim 1, further comprising:

a first etching stop layer disposed between the first interlayer insulating layer and the second interlayer insulating layer; and

a second etching stop layer disposed between the second interlayer insulating layer and the third interlayer insulating layer.

6. The semiconductor device of claim 1, wherein at least a part of the first wiring filling layer is disposed in the first via trench.

7. The semiconductor device of claim 1, wherein the first wiring filling layer is formed of a single film.

8. The semiconductor device of claim 1, wherein the second wiring pattern is spaced apart from the first via in a second direction crossing the first direction.

9. The semiconductor device of claim 1, further comprising:

a fourth wiring pattern disposed in a fourth trench formed in the first interlayer insulating layer and spaced apart from the first wiring pattern in a second direction crossing the first direction;

a second via trench extending from an upper surface of the fourth wiring pattern to a bottom surface of the second trench in the first direction; and

a second via including a second via barrier layer and a second via filling layer, wherein the second via barrier layer is disposed along side walls of the second via trench, and the second via filling layer fills the second via trench and is disposed on the second via barrier layer,

wherein the second wiring pattern includes a second wiring barrier layer and a second wiring filling layer,

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wherein the second wiring barrier layer is disposed along side walls of the second trench, and the second wiring filling layer fills the second trench and is disposed on the second wiring barrier layer,

wherein a bottom surface of the second wiring filling layer is in contact with the second interlayer insulating layer,

wherein the second via filling layer is in contact with each of the fourth wiring pattern and the second wiring filling layer, and

wherein the second wiring filling layer and the second via filling layer include different materials from each other.

10. The semiconductor device of claim 9, wherein the second wiring filling layer comprises

a second liner layer disposed along side walls of the second wiring barrier layer and the bottom surface of the second trench; and

a second filling layer filling the second trench and disposed on the second liner layer and the second via filling layer.

11. The semiconductor device of claim 9, wherein the second via filling layer includes:

a first portion disposed in the second via trench; and

a second portion protruding into the second trench and connected to the first portion of the second via filling layer.

12. A semiconductor device comprising:

a substrate;

a first interlayer insulating layer disposed on the substrate;

a first wiring pattern disposed in a first trench formed in the first interlayer insulating layer;

a first etching stop layer disposed on the first interlayer insulating layer;

a second interlayer insulating layer disposed on the first etching stop layer;

a second etching stop layer disposed on the second interlayer insulating layer;

a third interlayer insulating layer disposed on the second etching stop layer;

a second wiring pattern disposed in a second trench formed in the third interlayer insulating layer, and including a wiring barrier layer, a liner layer, and a filling layer, wherein the wiring barrier layer is disposed along side walls of the second trench, wherein the liner layer is disposed along side walls of the wiring barrier layer and a bottom surface of the second trench, wherein the filling layer fills the second trench and is disposed on the liner layer, and wherein the liner layer is in contact with the third interlayer insulating layer;

a via trench extending from an upper surface of the first wiring pattern to the bottom surface of the second trench in a vertical direction; and

a via including a via barrier layer and a via filling layer, wherein the via barrier layer is disposed along side walls of the via trench, wherein the via filling layer fills the via trench and is disposed on the via barrier layer, and wherein the via filling layer is in contact with each of the first wiring pattern and the filling layer,

wherein the filling layer and the via filling layer include different materials from each other.

13. The semiconductor device of claim 12, wherein the via filling layer includes:

a first portion disposed in the via trench; and

a second portion protruding into the second trench and connected to the first portion of the via trench.

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14. The semiconductor device of claim 13, wherein at least a part of a bottom surface of the second portion of the via filling layer is in contact with the third interlayer insulating layer.

15. The semiconductor device of claim 12, wherein an uppermost surface of the via barrier layer is in contact with the liner layer.

16. The semiconductor device of claim 12, further comprising:

a third wiring pattern disposed in a third trench formed in the second interlayer insulating layer and spaced apart from the via in a horizontal direction.

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